Muhammad Ammar

Junior AI/ML Engineer

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PROFILE

Computer Science graduate (2026) with hands-on experience in applied machine learning, deep learning, and computer vision. Developed a real-time, two-stage defect-detection system using ONNX and TensorRT optimization, with primary responsibility for the quality-control component of a team graduation project. Also completed classification and regression projects with FastAPI interfaces during a project-based ML internship. Seeking AI/ML internship or junior-level roles.

PROJECTS

- **Real-Time Industrial Quality Control System**: **Team Graduation Project**

  Stack: YOLO26n, PyTorch, OpenCV, ONNX, TensorRT, FastAPI, Firebase

  - Led the quality-control component, developing a real-time **two-stage YOLO26n pipeline** for product localization and packaging defect detection.
  - Achieved **99.4% mAP50** for localization and **94.2% mAP50** for defect detection; collected additional footage and collaborated on annotation.
  - Optimized inference using **ONNX** and **TensorRT**, increasing pipeline throughput on an RTX 3050 (4 GB) from **15–20 FPS with PyTorch** to **50 FPS with ONNX** and **140–170 FPS with TensorRT**.
  - Implemented part of the **FastAPI** streaming service and integrated **Firebase** for defect-event storage, collaborating with the team on the monitoring dashboard.

- **Road Accident Risk Prediction**

  Stack: Python, Scikit-learn, HistGradientBoosting, XGBoost

  - Built an end-to-end regression pipeline on **517K+ records**, comparing Linear Regression, Gradient Boosting, HistGradientBoosting, AdaBoost, and XGBoost.
  - Improved **test R^2 from 0.804 (Linear Regression baseline) to 0.887** using HistGradientBoosting, one-hot encoding, and engineered numerical and categorical features; tuned with **RandomizedSearchCV (5-fold cross-validation)**.
  - Serialized the model and preprocessing artifacts for reusable batch and single-record inference, with documented training and prediction scripts.

- **AI Body Measurement System**

  Stack: TensorFlow, Keras, Computer Vision, FastAPI, Python

  - Designed a **dual-branch CNN regression model** that fuses front and side body silhouettes with gender, height, and weight to predict **14 body measurements**.
  - Achieved a best observed **validation MAE of 2.86 cm** across the 14 measurements during training, using regularization, learning-rate scheduling, and early stopping.
  - Built an inference pipeline using **rembg** for background removal and **FastAPI** for model serving, integrating rule-based clothing-size recommendations through a browser-based interface.

- **NYC Taxi Trip Duration Prediction**

  Stack: Python, Scikit-learn, Pandas, NumPy, FastAPI

  - Cleaned **925K+ training records** using IQR-based outlier filtering, duplicate removal, and zero-distance filtering; engineered **Haversine distance** and time-based features.
  - Built a regression pipeline combining **degree-3 polynomial features**, MinMax scaling, and Ridge Regression, achieving a held-out **R^2 of 58.8%** on the log-transformed trip-duration target.
  - Deployed the model via **FastAPI**, reusing preprocessing and converting predictions to minutes.

EXPERIENCE

- **Machine Learning Intern**, Cellula Technology — Oct 2025 – Dec 2025 (Project-based)
  - **Hotel Booking Cancellation Prediction:** Built a classification pipeline on **500K+** hotel booking records, achieving **87% accuracy** and **0.91 F1-score** on the minority class.
  - **Uber Fare Prediction:** Built a regression pipeline on **1M+** trip records using gradient boosting, achieving R^2 **of 0.74** and **MAE of 1.01**.
  - Developed both projects end-to-end, covering **data cleaning, EDA, feature engineering, model training, and evaluation**, and served each model through a **FastAPI REST API** with a lightweight HTML interface for interactive local predictions.

SKILLS

- **Machine Learning & Deep Learning:** Supervised Learning, Classification, Regression, CNNs, RNNs, Feature Engineering, Model Evaluation, Hyperparameter Tuning
- **Computer Vision:** Object Detection, YOLO, OpenCV, MediaPipe
- **Frameworks & Libraries:** PyTorch, TensorFlow, Keras, Scikit-learn, XGBoost
- **Data Analysis & Scientific Python:** Python, NumPy, Pandas, Matplotlib, Seaborn
- **Databases:** SQL
- **Deployment & Inference:** FastAPI, ONNX, TensorRT, Firebase
- **Development Tools & DevOps:** Git, GitHub, Linux, Docker

COURSES & CERTIFICATIONS

- **deeplearning.ai:** Machine Learning Specialization · Deep Learning Specialization · Machine Learning in Production
- **CSkilled:** Machine Learning Diploma

EDUCATION

Mansoura University | Faculty of Computer and Information Science — 2022 – 2026

Bachelor's Degree in Computer Science

LANGUAGES

English, Arabic